

# Soc 596: Computational Social Science

Matthew J. Salganik  
Department of Sociology  
Princeton University

01-01-Introduction to computational social science



What is computational social science?

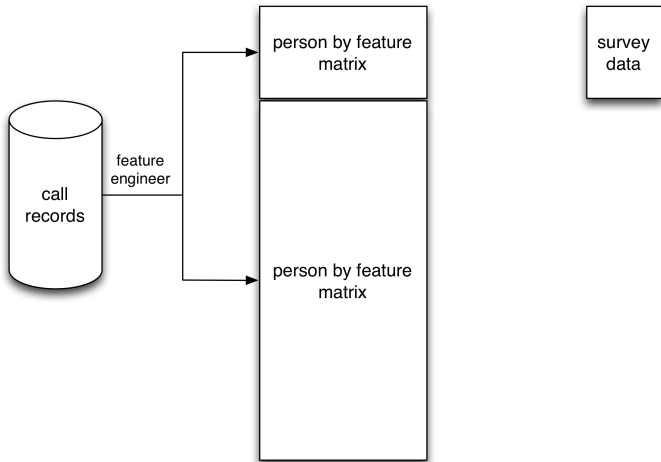
# Predicting poverty and wealth from mobile phone metadata

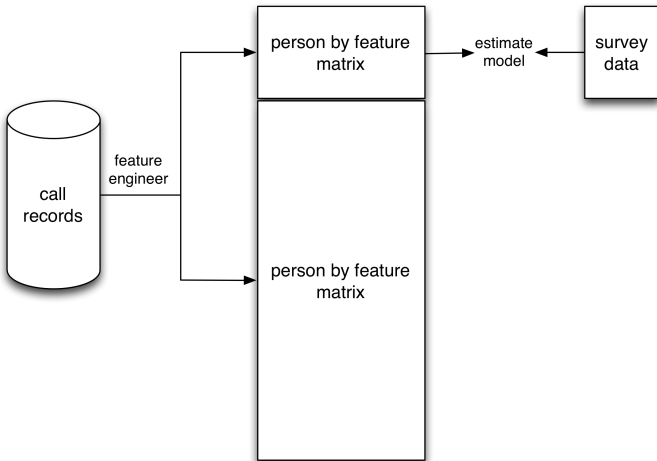
Joshua Blumenstock,<sup>1\*</sup> Gabriel Cadamuro,<sup>2</sup> Robert On<sup>3</sup>

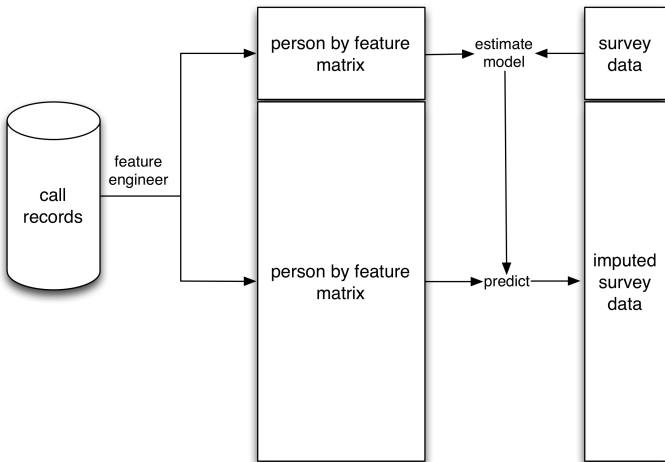
<http://dx.doi.org/10.1126/science.aac4420>

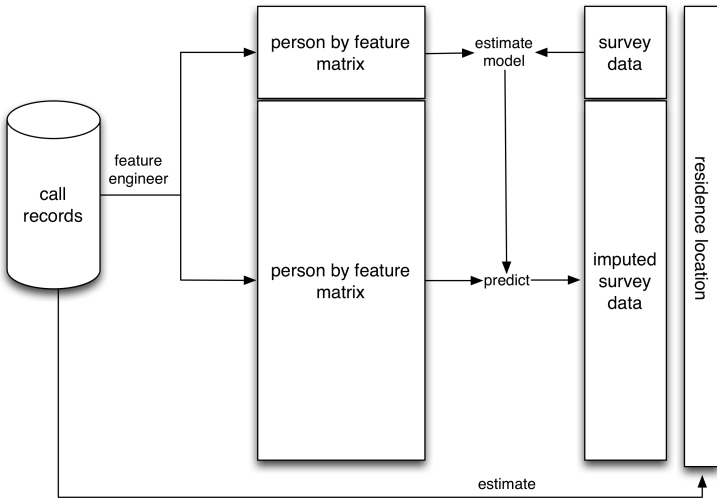


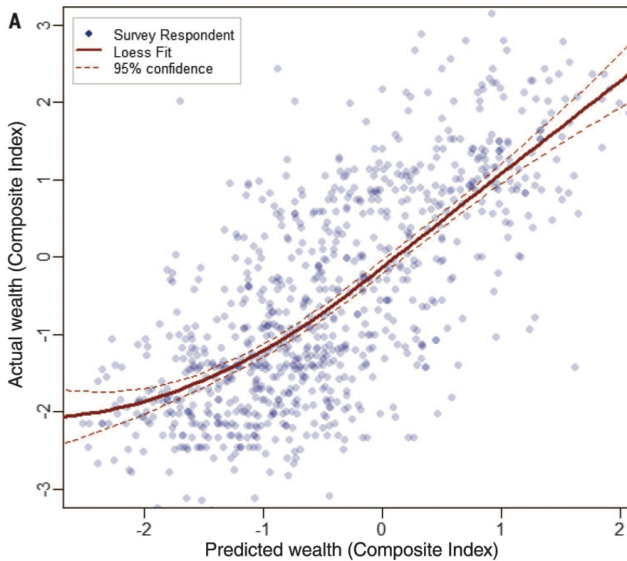




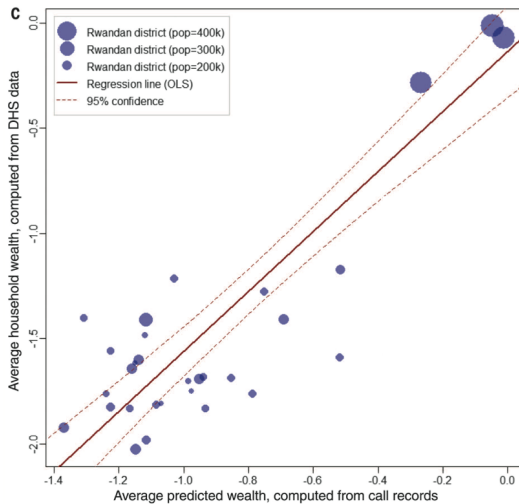












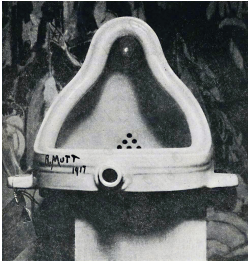
- ▶ 10 times faster
- ▶ 50 times cheaper

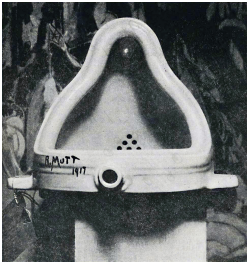
- ▶ *computational and social science*

- ▶ *computational* and *social science*
- ▶ often involves ethical/privacy questions that are now considered complex

- ▶ *computational* and *social science*
- ▶ often involves ethical/privacy questions that are now considered complex
- ▶ combines readymades and custommades







Readymades



Custommade

[https://commons.wikimedia.org/wiki/File:Duchamp\\_Fontaine.jpg](https://commons.wikimedia.org/wiki/File:Duchamp_Fontaine.jpg)  
[https://commons.wikimedia.org/wiki/File:%27David%27\\_by\\_Michelangelo\\_JBU0001.JPG](https://commons.wikimedia.org/wiki/File:%27David%27_by_Michelangelo_JBU0001.JPG)

- ▶ *computational and social science*
- ▶ often involves ethical/privacy questions that are now considered complex
- ▶ combines readymades and custommades

- ▶ *computational and social science*
- ▶ often involves ethical/privacy questions that are now considered complex
- ▶ combines readymades and custommades
- ▶ involves five key communities: social science, data science, business people, privacy advocates, policy makers

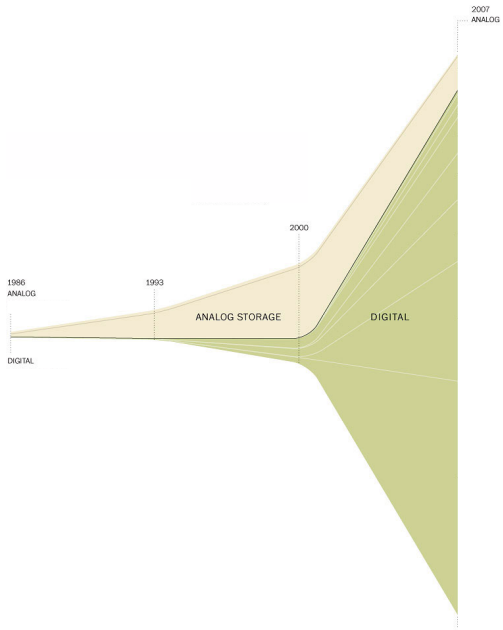
Balance between social science, data science, business people,  
privacy advocates, and policy makers

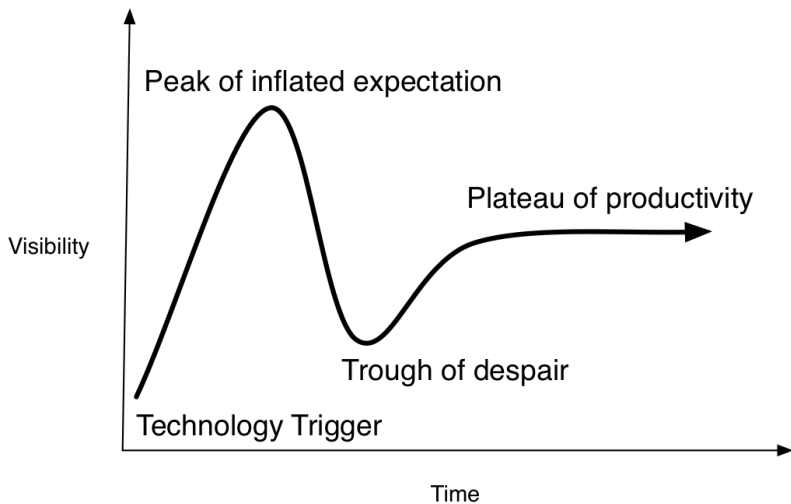
Health Ecosystem vs Horrific Monoculture

# What is computational social science?

- ▶ *computational and social science*
- ▶ often involves ethical/privacy questions that are now considered complex
- ▶ combines readymades and custommades
- ▶ involves five key communities: social science, data science, business people, privacy advocates, policy makers

Isn't computational social science a fad?





[https://commons.wikimedia.org/wiki/File:Gartner\\_Hype\\_Cycle.svg](https://commons.wikimedia.org/wiki/File:Gartner_Hype_Cycle.svg)

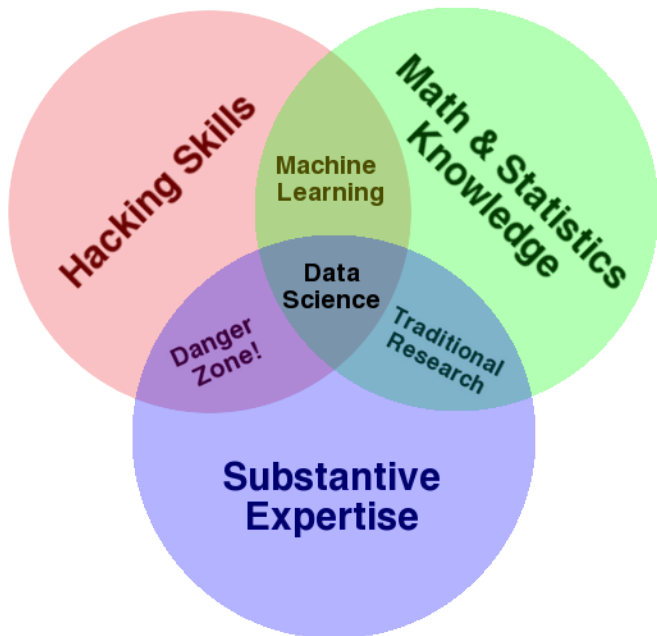
# Isn't computational social science a fad?

No

How do we create computational social science, individually and as a community?

Social Scientists  $\longleftrightarrow$  Data Scientists

# What is data science?



# What is data science?

Academic data science and industrial data science

Academic data science (think of Donoho article)

- ▶ goes back to Tukey (1962)

Academic data science (think of Donoho article)

- ▶ goes back to Tukey (1962)
- ▶ learning from data

## Academic data science (think of Donoho article)

- ▶ goes back to Tukey (1962)
- ▶ learning from data
- ▶ it includes statistics and much more

Industrial data science (think of Provost and Fawcett)

- ▶ lots of engineering

## Industrial data science (think of Provost and Fawcett)

- ▶ lots of engineering
- ▶ certain kinds of problems—churn and pop-tarts—that are not as far from social science as they appear

Data science alone is not enough if we want to study social behavior (e.g., Anderson)

Data science alone is not enough if we want to study social behavior (e.g., Anderson)

Social science alone is not enough if we want to use new data sources

Wallach (2015) reports overhearing: “I don’t get it. Why is that research?”

Wallach (2015) reports overhearing: “I don’t get it. Why is that research?”

| Computer science | Social science      |
|------------------|---------------------|
| Study anything   | Study social things |
| Methods driven   | Question driven     |
| Large found data | Small designed data |
| Prediction       | Explanation         |

[http://videlectures.net/icml2015-wallach\\_social\\_science/](http://videlectures.net/icml2015-wallach_social_science/)

Another important difference between social science and data science:



Social Scientists  $\longleftrightarrow$  Data Scientists

*Computational* social science vs Computational *social science*

# Simplicity vs Complexity

My goals for this module:

- ▶ SWBAT *describe* computational social science
- ▶ SWBAT *assess* whether computational social science is a fad
- ▶ SWBAT *describe* the intellectual traditions that underly computational social science